Stat 286: Causal Inference with Applications, Fall 2025

Section 4: Linear Regression

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1 Overview

Recall that regressing Y on a single predictor X via

$$Y = \alpha + \beta X + \varepsilon, \qquad \mathbb{E}[\varepsilon] = 0,$$

estimates a *linear association* between X and Y, not necessarily a causal effect. The goal of this section is to interpret β causally,

2 Structural Linear Models

2.1 Setup

Let $i = 1, \ldots, n$ index units, $T_i \in \{0, 1\}$ be a binary treatment, and Y_i denote the outcome of interest. Throughout we are assuming SUTVA such that $Y_i = Y_i(T_i)$ and an i.i.d draw from a super-population $\{T_i, Y_i(0), Y_i(1)\}_{i=1}^n \sim \mathbb{P}^*$.

2.2 Potential Outcomes as Linear Models

Define

$$\alpha_i := Y_i(0), \qquad \beta_i := Y_i(1) - Y_i(0), \qquad \alpha := \mathbb{E}[\alpha_i] = \mathbb{E}[Y_i(0)], \qquad \beta := \mathbb{E}[\beta_i] = \mathbb{E}[Y_i(1) - Y_i(0)].$$

Then, for $t \in \{0, 1\}$,

$$\begin{aligned}
Y_i(t) &= Y_i(0) + \big(Y_i(1) - Y_i(0)\big)t \\
&= \mathbb{E}[Y_i(0)] + \big(Y_i(0) - \mathbb{E}[Y_i(0)]\big) + \mathbb{E}[Y_i(1) - Y_i(0)]\,t + \big(Y_i(1) - Y_i(0) - \mathbb{E}[Y_i(1) - Y_i(0)]\big)\,t \\
&= \alpha + (\alpha_i - \alpha) + \beta\,t + (\beta_i - \beta)\,t \\
&= \alpha + \beta\,t + \underbrace{(\beta_i - \beta)\,t + (\alpha_i - \alpha)}_{=:\ \varepsilon_i(t)} \\
&= \alpha + \beta\,t + \varepsilon_i(t).
\end{aligned}$$

By construction, $\mathbb{E}[\varepsilon_i(t)] = 0$ for each $t \in \{0, 1\}$. Hence, *without* assuming any parametric assumptions we can write the potential outcomes in a linear form:

$$Y_i(t) \;=\; \alpha + \beta t + \varepsilon_i(t), \qquad \mathbb{E}[\varepsilon_i(t)] = 0, \tag{1}$$

where the error term depends on t and is thus called the *heterogeneous* linear model. Thus regressing $Y_i \sim [1, T_i]$ targets $\beta = \mathbb{E}[Y_i(1) - Y_i(0)]$.

Beyond binary T. For multi-valued *categorical* T, we can introduce indicators and proceed analogously without imposing further assumptions. For a *continuous* T, writing $Y_i(t) = \alpha_i + \beta_i t$ does impose linearity in t (constant marginal effect $\partial Y_i(t)/\partial t = \beta_i$), which is a parametric assumption.

2.3 The Homogeneous Model

If we compare the above with the *homogeneous* linear model

$$Y_i(t) \;=\; \alpha + \beta t + \varepsilon_i, \qquad t \in \{0,1\}, \qquad \mathbb{E}[\varepsilon_i] = 0,$$

with the error term independent of t, then the homogeneous model implies an *constant additive* effect: $Y_i(1) - Y_i(0) = \beta$ for all $i = 1, \dots, n$. This is a strong assumption (compare with the sharp null from earlier chapters) and underlines the importance of allowing the error term to depend on the treatment. However, the homogeneous and heterogeneous specifications share the same ATE, $\beta = \mathbb{E}[\beta_i]$, and are observationally indistinguishable from the data.

2.4 Why Linear Regression?

The advantage of viewing potential outcomes as linear models is that we can easily generalize them to incorporate covariates, for example,

$$Y_i \;=\; \alpha \;+\; \beta T_i \;+\; \boldsymbol{\gamma}^\top \boldsymbol{X}_i \;+\; \varepsilon_i.$$

In experimental studies this can decrease the variance of our estimates and in observational studies this can make the strong ignorability assumption more believable. In this module, we study the experimental setup. We start with a review of the most important linear regression facts.

3 Linear Regression Review

3.1 The Linear Model

Consider the linear model

$$\boldsymbol{Y} = \boldsymbol{X}\boldsymbol{\beta}^* + \boldsymbol{\varepsilon},$$

where

$$\boldsymbol{Y} = \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix}, \quad \boldsymbol{X} = \begin{bmatrix} 1 & X_{11} & \cdots & X_{K1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n1} & \cdots & X_{Kn} \end{bmatrix}, \quad \boldsymbol{\beta}^* = \begin{bmatrix} \beta_0^* \\ \vdots \\ \beta_K^* \end{bmatrix}.$$

Since we have not specified $\boldsymbol{\varepsilon}$, this model (so far) does not impose any parametric assumption. Traditionally, estimation is done with the Ordinary Least Squares (OLS) estimator:

$$\hat{\boldsymbol{\beta}} = \operatorname*{argmin}_{\boldsymbol{\beta}} \|\boldsymbol{Y} - \boldsymbol{X}\boldsymbol{\beta}\|_2,$$

which leads to the closed-form solution:

$$\hat{\boldsymbol{\beta}} = (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top \boldsymbol{Y} = \left(\frac{1}{n} \sum_{i=1}^n \boldsymbol{X}_i \boldsymbol{X}_i^\top \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n \boldsymbol{X}_i Y_i \right),$$

where $\boldsymbol{X}_i = [1, X_{1i}, \dots, X_{Ki}]^\top$. The OLS estimator gives the *best linear approximation* of $\mathbb{E}[Y_i \mid \boldsymbol{X}_i]$.

Assumption 1 (Strict exogeneity)

$$\mathbb{E}[\varepsilon_i \mid \boldsymbol{X}_i] = 0.$$

This assumption implies that the error term is *uncorrelated* with the covariates, $\text{Cov}(\boldsymbol{\varepsilon}, \boldsymbol{X}) = \boldsymbol{0}$. In a randomized design, this is indeed true, as the following lemma shows.

Lemma 1 (Randomized Assignment implies Strict Exogeneity) *Recall the linear model of Equation* (1). *Assuming randomized assignment,* $\{Y_i(0), Y_i(1)\} \perp T_i$ *for all* i*, we have*

$$\mathbb{E}[\epsilon_i(T_i) \mid T_i] = 0$$

Proof: Recall

$$\epsilon_i(T_i) = (\beta_i - \beta)T_i + (\alpha_i - \alpha) = ((Y_i(1) - Y_i(0)) - \mathbb{E}[Y_i(1) - Y_i(0)])T_i + (Y_i(0) - \mathbb{E}[Y_i(0)])$$

such that under randomization and i.i.d. sampling

$$\mathbb{E}[\epsilon_i(T_i) \mid T_i] = (\mathbb{E}[Y_i(1) - Y_i(0)] - \mathbb{E}[Y_i(1) - Y_i(0)])T_i + \mathbb{E}[Y_i(0)] - \mathbb{E}[Y_i(0)] = 0.$$

□

However, as we will see, if covariates are included, strict exogeneity does not necessarily hold under a CRD. Instead, it imposes a *linearity* assumption.

Lemma 2 *Assuming strict exogeneity, then the following holds*

- ***Unbiasedness.***

$$\mathbb{E}[\hat{\boldsymbol{\beta}} \mid \boldsymbol{X}] = \boldsymbol{\beta}^*.$$

- ***Finite-sample variance (sandwich form).***

$$\mathrm{Var}(\hat{\boldsymbol{\beta}} \mid \boldsymbol{X}) = \underbrace{(\boldsymbol{X}^\top \boldsymbol{X})^{-1}}_{\textit{Bread}} \underbrace{\boldsymbol{X}^\top \boldsymbol{\Omega} \boldsymbol{X}}_{\textit{Meat}} \underbrace{(\boldsymbol{X}^\top \boldsymbol{X})^{-1}}_{\textit{Bread}},$$

where

$$\boldsymbol{\Omega} = \mathrm{diag}(\sigma_1^2, \dots, \sigma_n^2), \qquad \sigma_i^2 = \mathrm{Var}(\varepsilon_i \mid \boldsymbol{X}_i).$$

This is the foundation for all robust (EHW/HC2) and homoskedastic OLS variance formulas.

3.2 Variance

3.2.1 Homoskedasticity

Typically in OLS, we assume that the variance is homoskedastic.

Assumption 2 (Homoskedasticity) *Errors have constant variance*

$$\mathrm{Var}(\varepsilon_i \mid \boldsymbol{X}_i) = \sigma^2.$$

Assumption 2 then recovers the well known variance formula.

$$\mathrm{Var}(\hat{\boldsymbol{\beta}} \mid \boldsymbol{X}) = (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top \boldsymbol{\Omega} \boldsymbol{X} (\boldsymbol{X}^\top \boldsymbol{X})^{-1} = \sigma^2 (\boldsymbol{X}^\top \boldsymbol{X})^{-1}.$$

However, this assumption os often unrealistic in practice in causal inference application, as it assumes that the variance in treatment and control is equal. Often, the variance of treatment exceeds the variance in control, as the treatment injects additional noise. For example, giving a new drug to patients may have many unintended consequences on the outcome compared to giving a placebo.

Example 1 *Recall the model in Equation* (1) *and let* $\boldsymbol{X}_i = [1, T_i]^\top$. *Then under CRD we have*

$$(\boldsymbol{X}^\top \boldsymbol{X})^{-1} = \left(\begin{bmatrix} \sum_i 1 & \sum_i T_i \\ \sum_i T_i & \sum_i T_i^2 \end{bmatrix} \right)^{-1} = \frac{1}{n_1 n_0} \begin{bmatrix} n_1 & -n_1 \\ -n_1 & n \end{bmatrix}.$$

Thus,

$$(\boldsymbol{X}^\top \boldsymbol{X})^{-1}_{22} = \frac{n}{n_1 n_0} = \frac{1}{\sum_{i=1}^n (T_i - \overline{T})^2}$$

under CRD. Assuming that $\mathrm{Var}(\epsilon_i(0)) = \mathrm{Var}(Y(1)) = \mathrm{Var}(Y(0)) = \mathrm{Var}(\epsilon_i(1))$ *(equal variance of treatment and control), which by random sampling implies (by Lemma 1)* $\mathrm{Var}(\epsilon_i) = \mathrm{Var}(\epsilon_j) = \sigma^2$ *that*

$$\mathrm{Var}(\hat{\beta}_T \mid \boldsymbol{X}) = \sigma^2 \left(\frac{1}{n_1} + \frac{1}{n_0} \right).$$

The typical plug-in estimator is given by

$$\widehat{\mathrm{Var}}(\hat{\beta}_T \mid T) \;=\; \frac{\hat{\sigma}^2}{\sum_{i=1}^n (T_i - \overline{T})^2}, \qquad \hat{\sigma}^2 \;=\; \frac{1}{n-2}\sum_{i=1}^n \hat{\varepsilon}_i^2.$$

However, if homoskedasticity does not hold and $\sigma_1^2 = \mathrm{Var}(Y(1)) \neq \mathrm{Var}(Y(0)) = \sigma_0^2$ *then the true Neymann variance is*

$$\frac{\sigma_1^2}{n_1} + \frac{\sigma_0^2}{n_0}.$$

Hence, we obtain a bias

$$\mathrm{Bias} \;=\; \mathbb{E}\left[\frac{\hat{\sigma}^2}{\sum_{i=1}^n (T_i - \overline{T})^2}\right] - \left(\frac{\sigma_1^2}{n_1} + \frac{\sigma_0^2}{n_0}\right) \;=\; \frac{(n_1 - n_0)(n-1)}{n_1 n_0 (n-2)}\,(\sigma_1^2 - \sigma_0^2),$$

which vanishes if either $n_1 = n_0$ *(balanced design) or* $\sigma_1^2 = \sigma_0^2$ *(homoskedastic). The bias may be positive or negative, so it is not uniformly conservative and hence this variance estimator cannot be used in practice.*

3.2.2 Heteroskedasticity: Eicker–Huber–White (EHW, HC0)

As we have seen, it is important that the variance is allowed to vary across units. One estimator that takes that into account is the Eicker–Huber–White (EHW or HC0) estimator, which is given by

$$\widehat{\mathrm{Var}}_{\mathrm{HC0}}(\hat{\boldsymbol{\beta}} \mid \boldsymbol{X}) = (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top \,\mathrm{diag}(\hat{\varepsilon}_i^2)\, \boldsymbol{X} (\boldsymbol{X}^\top \boldsymbol{X})^{-1},$$

where $\hat{\varepsilon}_i = Y_i - \boldsymbol{X}_i^\top \hat{\boldsymbol{\beta}}$. We compare this again to our example of regression $Y_i \sim [1, T_i]$

Example 2 *Recall the model in Equation* (1) *and let* $\boldsymbol{X}_i = [1, T_i]^\top$. *Then using the HC0 variance estimator we obtain*

$$\widehat{\mathrm{Var}}_{\mathrm{HC0}}(\hat{\beta}_T \mid \boldsymbol{X}) = \frac{\tilde{\sigma}_1^2}{n_1} + \frac{\tilde{\sigma}_0^2}{n_0},$$

where

$$\tilde{\sigma}_t^2 = \frac{1}{n_t} \sum_{i:T_i = t} \left(Y_i - \overline{Y}_t\right)^2.$$

However, this underestimates the variance. Indeed, following Neyman's correction, we would like to have

$$\widehat{\mathrm{Var}}(\hat{\beta}_T \mid \boldsymbol{X}) = \frac{\hat{\sigma}_1^2}{n_1} + \frac{\hat{\sigma}_0^2}{n_0}.$$

where

$$\hat{\sigma}_t^2 = \frac{1}{n_t - 1} \sum_{i:T_i = t} \left(Y_i - \overline{Y}_t\right)^2.$$

Thus, in this case, HC0 is consistent *but biased in finite samples.*

3.2.3 Heteroskedasticity: HC2

The idea of the HC0 estimator is to correct HC0's downward bias using leverage adjustment.

$$\widehat{\mathrm{Var}}_{\mathrm{HC2}}(\hat{\boldsymbol{\beta}} \mid \boldsymbol{X}) = (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top \,\mathrm{diag}\left(\frac{\hat{\varepsilon}_i^2}{1 - p_{ii}}\right) \boldsymbol{X} (\boldsymbol{X}^\top \boldsymbol{X})^{-1},$$

where $p_{ii} = \boldsymbol{X}_i^\top (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}_i$ is the leverage of observation i.

Example 3 *Recall the model in Equation (1) and let $\boldsymbol{X}_i = [1, T_i]^\top$.*

$$p_{ii} = \frac{1}{n_1 n_0} \begin{bmatrix} 1 \\ T_i \end{bmatrix}^\top \begin{bmatrix} n_1 & -n_1 \\ -n_1 & n \end{bmatrix} \begin{bmatrix} 1 \\ T_i \end{bmatrix} = \frac{1}{n_1} T_i + \frac{1}{n_0}(1 - T_i).$$

Thus

$$\frac{1}{1 - p_{ii}} = \begin{cases} \frac{n_1}{n_1 - 1}, & T_i = 1, \\ \frac{n_0}{n_0 - 1}, & T_i = 0. \end{cases}$$

This correction yields the Neyman variance:

$$\widehat{\mathrm{Var}}_{\mathrm{HC2}}(\hat{\beta}_T \mid \boldsymbol{X}) = \frac{\hat{\sigma}_1^2}{n_1} + \frac{\hat{\sigma}_0^2}{n_0}, \quad \hat{\sigma}_t^2 = \frac{1}{n_t - 1} \sum_{i: T_i = t} \left(Y_i - \overline{Y}_t\right)^2.$$

In very small samples, adjust further by scaling with $\frac{n}{n-(K+1)}$ *or use* HC3.

4 Linear Models in Experiments

It is worth to recall our assumptions which are valid throughout this section. Let $i = 1, \ldots, n$ index units, $T_i \in \{0, 1\}$ be a binary treatment, and Y_i denote the outcome of interest. Throughout we are assuming SUTVA such that $Y_i = Y_i(T_i)$ and an i.i.d draw from a super-population $\{T_i, Y_i(0), Y_i(1)\}_{i=1}^n \sim \mathbb{P}^*$. We further assume a CRD such that $\{Y_i(0), Y_i(1)\} \perp T_i$ for all i.

4.1 Linear Regression with No Covariates

Recall the model in Equation (1),

$$Y_i(t) \;=\; \alpha + \beta t + \varepsilon_i(t).$$

This suggests regressing the observational data

$$Y_i \;=\; \alpha + \beta T_i + \varepsilon_i.$$

We already know by Lemma 1 that this will yield unbiased estimates $\mathbb{E}[\hat{\beta}] = \mathbb{E}[Y_i(1) - Y_i(0)]$. The next lemma shows that in this case we will recover the classic difference-in-means (DiM) estimator.

Lemma 3 *Let $\{(Y_i, T_i)\}_{i=1}^n$ with $T_i \in \{0, 1\}$. Then regressing $Y_i = \alpha + \beta T_i + \epsilon_i$ yields*

$$\hat{\alpha} = \frac{1}{n_0} \sum_{i=1}^n (1 - T_i) Y_i \;=\; \bar{Y}_0,$$
$$\hat{\beta} = \frac{1}{n_1} \sum_{i=1}^n T_i Y_i \;-\; \frac{1}{n_0} \sum_{i=1}^n (1 - T_i) Y_i \;=\; \hat{\tau},$$

where $n_1 = \sum_{i=1}^n, n_0 = n - n_1$.

Proof: Write the OLS slope as

$$\hat{\beta} = \frac{\sum_{i=1}^n (T_i - \bar{T})(Y_i - \bar{Y})}{\sum_{i=1}^n (T_i - \bar{T})^2}.$$

Denominator. Since $T_i \in \{0, 1\}$,

$$\sum_{i=1}^n (T_i - \bar{T})^2 = n_1 (1 - \bar{T})^2 + n_0 (0 - \bar{T})^2 = \frac{n_1 n_0}{n}.$$

Numerator. Because $\sum_i (Y_i - \bar{Y}) = 0$,

$$\sum_{i=1}^{n}(T_i - \bar{T})(Y_i - \bar{Y}) = \sum_{i=1}^{n} T_i(Y_i - \bar{Y}) = \sum_{i:T_i=1}(Y_i - \bar{Y}) = n_1(\bar{Y}_1 - \bar{Y}).$$

With $\bar{Y} = (n_1\bar{Y}_1 + n_0\bar{Y}_0)/n$ we have

$$\bar{Y}_1 - \bar{Y} = \bar{Y}_1 - \frac{n_1\bar{Y}_1 + n_0\bar{Y}_0}{n} = \frac{n_0}{n}\,(\bar{Y}_1 - \bar{Y}_0).$$

Hence the numerator equals $\frac{n_1 n_0}{n}(\bar{Y}_1 - \bar{Y}_0)$, and dividing by the denominator gives

$$\hat{\beta} = \bar{Y}_1 - \bar{Y}_0.$$

Finally,

$$\hat{\alpha} = \bar{Y} - \hat{\beta}\,\bar{T} = \frac{n_1\bar{Y}_1 + n_0\bar{Y}_0}{n} - \frac{n_1}{n}(\bar{Y}_1 - \bar{Y}_0) = \bar{Y}_0.$$

□

Using the HC2 variance estimator recovers,

$$\widehat{\mathrm{Var}}_{\mathrm{HC2}}(\hat{\beta}_{\mathrm{DiM}} \mid \boldsymbol{T}) = \frac{\hat{\sigma}_1^2}{n_1} + \frac{\hat{\sigma}_0^2}{n_0}, \quad \hat{\sigma}_t^2 = \frac{1}{n_t - 1}\sum_{i:T_i=t}(Y_i - \overline{Y}_t)^2.$$

Thus, OLS with an intercept and binary T_i using HC2 variance estimation reproduces the classical difference-in-means (DiM) estimator with Neymann variance.

4.2 Linear Regression with Covariates (Pooled)

Even under randomization, covariates can be imbalanced between treatment and control; adjusting for X can improve precision and lower the variance. This suggest considering the *pooled* (ANCOVA) model

$$Y_i \;=\; \alpha \;+\; \beta\,T_i \;+\; \boldsymbol{\gamma}^{\top}\boldsymbol{X}_i \;+\; \varepsilon_i. \tag{2}$$

Note that since we include covariates, even under CRD, exogeneity might not hold. Instead, the model in Equation (2) assumes a *linear* effect of the covariates on the outcomes, which will also bias our estimate for β.

This model is called the pooled model, as it assumes that the slopes between treatment and control are the same (common $\boldsymbol{\gamma}$)

$$\begin{aligned} T_i = 1: \quad Y(1) \;&=\; \alpha + \beta \;+\; \boldsymbol{\gamma}^{\top}\boldsymbol{X}_i \;+\; \varepsilon_i, \\ T_i = 0: \quad Y(0) \;&=\; \alpha \;+\; \boldsymbol{\gamma}^{\top}\boldsymbol{X}_i \;+\; \varepsilon_i. \end{aligned}$$

Example 4 (Pooled model in the univariate case) *To get an intuition for the model in Equation (2) consider the simplified case where $X_i \in \mathbb{R}$. Then the normal equation of the OLS regressor imply*

$$\sum_{i=1}^{n}\hat{\varepsilon}_i = 0, \qquad \sum_{i=1}^{n} T_i\,\hat{\varepsilon}_i = 0, \qquad \sum_{i=1}^{n} X_i\,\hat{\varepsilon}_i = 0.$$

From the first two,

$$\bar{Y} \;=\; \hat{\alpha} + \hat{\beta}\,\frac{n_1}{n} + \hat{\gamma}\,\bar{X}, \qquad \bar{Y}_1 \;=\; \hat{\alpha} + \hat{\beta} + \hat{\gamma}\,\bar{X}_1.$$

(Equivalently, using $\sum(1 - T_i)\hat{\varepsilon}_i = 0$: $\bar{Y}_0 = \hat{\alpha} + \hat{\gamma}\,\bar{X}_0$.) Eliminating $\hat{\alpha}$ yields

$$\hat{\beta} \;=\; (\bar{Y}_1 - \bar{Y}_0) \;-\; \hat{\gamma}\,(\bar{X}_1 - \bar{X}_0) \;=\; \hat{\beta}_{DiM} \;-\; \hat{\gamma}\,(\bar{X}_1 - \bar{X}_0).$$

Thus this model yields a bias

$$\mathbb{E}[\hat{\beta}] \;=\; \mathbb{E}\big[Y(1) - Y(0)\big] \;-\; \underbrace{\mathbb{E}\big[\hat{\gamma}\,(\bar{X}_1 - \bar{X}_0)\big]}_{\neq 0,\mathrm{Cov}(\hat{\gamma},\bar{X}_1-\bar{X}_0)\neq 0,\hat{\gamma} \textit{ depends on } (X_0,X_1)} \;\neq \mathbb{E}\big[Y(1) - Y(0)\big].$$

Under random sampling and CRD, $\bar{X}_1 - \bar{X}_0 = O_p(n^{-1/2})$ *so the bias vanishes as* $n \to \infty$.

Let's consider the variance of the estimator. If the model is correctly specified, $\mathbb{E}[Y \mid T, X] = \alpha + \beta T + \gamma X$ *and homoskedasticity holds, then*

$$\mathrm{Var}(\hat{\beta}_P \mid T, X) \;\approx\; \mathrm{Var}(\hat{\beta}_{DiM} \mid T, X)\,\big(1 - R^2_{Y\sim X}\big),$$

where $R^2_{Y\sim X}$ *is from regressing* Y *on* $(1, X)$*. Thus variance decreases with predictive power of* X*; without correct specification this reduction is not guaranteed.*

The intuition of Example 4 holds more generally. Indeed, we have the key properties of the pooled linear regression in experiments

- $\hat{\beta}_{\mathrm{P}} \to \beta$ (consistent), but not necessarily unbiased in finite samples.
- If $\boldsymbol{X}$ is predictive of Y, this regression can reduce variance.
- However, if the model is misspecified (e.g., treatment and control have different covariate effects), efficiency gains may be lost or even reversed.

4.3 Linear Regression with Covariates (Interacted)

The pooled model assumes equal slopes for treatment and control. To relax this, we allow interaction terms between treatment and covariates.

$$Y_i = \alpha + \beta T_i + \boldsymbol{\gamma}^\top \tilde{\boldsymbol{X}}_i + \boldsymbol{\delta}^\top (T_i \tilde{\boldsymbol{X}}_i) + \varepsilon_i,$$

where $\tilde{\boldsymbol{X}}_i = \boldsymbol{X}_i - \overline{\boldsymbol{X}}$ (demeaned covariates). Demeaning is important, as otherwise our estimates for β won't be consistent anymore. This issue was not present in the pooled case, as α can absorb the covariates mean. This model is called *interacted*, as now parallel slope assumption of the pooled model is relaxed

$$T_i = 1:\;\; Y(1) = \alpha + \beta + \boldsymbol{\delta}^\top \tilde{\boldsymbol{X}}_i + \varepsilon_i,$$
$$T_i = 0:\;\; Y(0) = \alpha + \boldsymbol{\gamma}^\top \tilde{\boldsymbol{X}}_i + \varepsilon_i.$$

Treatment and control now have different intercepts and covariate effects. The estimated treatment effect can be written as

$$\hat{\beta}_{\mathrm{I}} = \frac{1}{n} \sum_{i=1}^{n} \big[T_i(Y_i - \widehat{Y}_i(0)) + (\widehat{Y}_i(1) - Y_i)(1 - T_i)\big],$$

where

$$\widehat{Y}_i(t) = \hat{\alpha} + \hat{\beta}_{\mathrm{I}} t + \hat{\boldsymbol{\gamma}}^\top \tilde{\boldsymbol{X}}_i + \hat{\boldsymbol{\delta}}^\top (t \tilde{\boldsymbol{X}}_i).$$

This is also called the *imputation* form.

Under our assumptions, most importantly randomization, and minor regularity conditions this interacted model has some great properties, proved by Lin [2013].

1. **Consistency:**
$$\hat{\beta}_{\mathrm{I}} \;\xrightarrow{p}\; \tau = \mathbb{E}[Y_i(1) - Y_i(0)].$$
2. **Asymptotic normality:**
$$\sqrt{n}\,(\hat{\beta}_{\mathrm{I}} - \tau) \;\xrightarrow{d}\; \mathcal{N}\big(0,\, \sigma^2_{\mathrm{I}}\big)\,.$$
3. **Efficiency:**
$$\sigma^2_{\mathrm{I}} \;\leq\; \sigma^2_{\mathrm{DiM}} \quad \text{and} \quad \sigma^2_{\mathrm{I}} \;\leq\; \sigma^2_{\mathrm{P}}.$$
   That is, the interacted regression is at least as efficient as DiM or pooled OLS.
4. **Variance estimation:** The HC2 (heteroskedasticity-robust) variance estimator is consistent for $\hat{\beta}_{\mathrm{I}}$.
5. **Robustness:** These results hold even if the model is misspecified!

4.4 Linear Regression in Practice

Under randomization, we can follow the recipe described below

1. Center covariates: $\tilde{\boldsymbol{X}}_i = \boldsymbol{X}_i - \overline{\boldsymbol{X}}$.
2. Fit $Y \sim [1, T, \tilde{\boldsymbol{X}}, T\tilde{\boldsymbol{X}}]$. If n is small, fall back to DiM estimation.
3. Report $\hat{\beta}$ with HC2 (or HC3 for small n) and a 95% CI.

References

Winston Lin. Agnostic notes on regression adjustments to experimental data: Reexamining freedman's critique. *The Annals of Applied Statistics*, pages 295–318, 2013.